CS24303 — THEORY (3.0 CR) | MODULE I NOTES (14 TOPICS) | BIT MESRA | NEP SCHEME

Module 1: Data Attributes & Proximity Metrics

KDD Pipeline, 4 Attribute Types, Boxplots, Euclidean & Cosine Distance

Module I: Introduction to Data Mining — Complete 14-Topic Syllabus Tracker

Topic 1: Introduction to Data Mining (Knowledge Discovery Process)

Topic 2: Relational Databases (Tables, Tuples & Schemas)

Topic 3: Data Warehouses (SITN Characteristics & OLAP)

Topic 4: Transactional Databases (Transaction Sets & Market Basket)

Topic 5: Advanced Database Systems & Applications

Topic 6: Data Mining Functionalities (Characterization vs. Discrimination)

Topic 7: Classification of Data Mining Systems

Topic 8: Major Issues in Data Mining (Quality, Scalability, Privacy)

Topic 9: Data Concepts (Objects, Attributes & Datasets)

Topic 10: Data Objects & Attribute Types (Nominal, Binary, Ordinal, Numeric)

Topic 11: Basic Statistical Descriptions (Mean, Median, Mode, IQR, Box Plots)

Topic 12: Data Visualization (Histograms, Scatter Plots, Quantile Plots)

Topic 13: Measuring Data Similarity (Cosine Similarity & Dot Products)

Topic 14: Measuring Data Dissimilarity (Euclidean & Manhattan Distances)

Topic 1 & 2: Introduction to Data Mining & Relational Databases

Data Mining (Knowledge Discovery in Databases — KDD) is the computational process of discovering non-trivial, implicit, previously unknown, and potentially actionable patterns and models from massive datasets.

Database Query (SQL / OLTP)	Data Mining (KDD / Discovery)
Retrieves exact known records matching user-specified criteria (e.g., "Find all students with $\text{CGPA} > 8.0$").	Discovers hidden, inductive relationships and associations (e.g., "Students who score high in Algorithms also excel in Compiler Design").
Deductive, exact, deterministic SQL responses.	Inductive, statistical, pattern-based models with probabilistic confidence.

Topic 3 – 5: Data Warehouses & Advanced Databases [UPLOADED PYQ]

🧠 **Memory Hook: SITN Characteristics of Data Warehouses (Inmon)**

- **S — Subject-Oriented:** Organized around major business subjects (Customers, Products, Sales) rather than transactional operations.
- **I — Integrated:** Data from diverse operational sources is standardized into consistent naming, units, and encoding schemes.
- **T — Time-Variant:** Maintains historical data spanning 5–10 years with explicit timestamp attributes for trend analysis.
- **N — Non-Volatile:** Read-only analytical storage; historical snapshots are loaded and not updated in-place.

Operational Database (OLTP)	Data Warehouse (OLAP) [UPLOADED PYQ — IT335 Mid]
Supports day-to-day operational transactions (e.g., bank ATM withdrawal).	Supports long-term executive decision making, reporting, and predictive analytics.
Stores current, real-time up-to-the-second data.	Stores historical snapshots over long multi-year horizons.
Highly normalized tables (3NF/BCNF) to eliminate update anomalies.	Denormalized dimensional schemas (Star / Snowflake) to optimize complex query joins.

Topic 6: Data Mining Functionalities [UPLOADED PYQ]

Functionality	Core Algorithmic Objective	Illustrative Example
1. Characterization [UPLOADED PYQ]	Summarizes the general properties of a target class into compact descriptions (Attribute-Oriented Induction).	"Summarize the demographic profile of customers who spent $> ₹1,00,000$ last year."
2. Discrimination [UPLOADED PYQ]	Compares the target class against one or a set of contrasting comparative classes.	"Compare customers who buy gaming laptops vs. customers who buy standard business laptops."
3. Association & Correlation	Finds frequent itemsets and conditional dependency rules ($X \implies Y$).	"Customers purchasing Milk and Bread also buy Butter with 80% confidence."
4. Classification	Supervised learning predicting discrete categorical class labels.	"Classifying credit card transactions as Fraudulent or Legitimate."
5. Prediction (Regression)	Predicts continuous real-valued numeric values.	"Predicting real-estate house prices based on square footage and location."
6. Clustering	Unsupervised grouping of unlabeled data points based on geometric similarity.	"Customer market segmentation into 5 distinct spending behavioral clusters."
7. Outlier Analysis	Detects anomalous objects that deviate significantly from normal baseline distributions.	"Network intrusion detection and credit card fraud surveillance."

Topic 8 – 10: Data Objects & Attribute Types

Attribute Type	Mathematical Characteristics & Valid Operations	Representative Examples
1. Nominal	Categorical labels, names, or codes with **no meaningful rank or ordering** ($=, \neq$).	`Marital_Status` (Single, Married), `Color` (Red, Blue).
2. Binary	Nominal attribute with exactly 2 states (`0` and `1`). Symmetric (Gender) vs. Asymmetric (Medical test positive/negative).	`Smoker` (Yes/No), `COVID_Positive` (0/1).
3. Ordinal	Categorical values with a **meaningful ranking / order**, but intervals between values cannot be measured ($<, >, =, \neq$).	`Customer_Rating` (Poor < Fair < Good < Excellent), `Academic_Grade` (A, B, C, D, F).
4. Numeric (Interval)	Measured on a scale of equal units; **zero is arbitrary** (no true zero point; addition and subtraction meaningful, ratios meaningless).	Temperature in $^\circ\mathrm{C}$ or $^\circ\mathrm{F}$, Calendar Years.
5. Numeric (Ratio)	Inherent **absolute zero point** (both differences and ratios are mathematically meaningful: $y = 2x$).	`Salary` (₹$50,000$), `Age` (20 years), `Weight`, `Length`.

Topic 11 & 12: Basic Statistical Descriptions & Visualization

Five-Number Summary & Box-Plot Formulations:

$$\text{Five-Number Summary} = \langle \text{Minimum},\ Q_1\ (25\%),\ \text{Median}\ (Q_2),\ Q_3\ (75\%),\ \text{Maximum} \rangle$$

$$\text{Interquartile Range (IQR)} = Q_3 - Q_1$$

$$\text{Outlier Cutoff Boundaries: Lower} = Q_1 - 1.5 \times \text{IQR}, \quad \text{Upper} = Q_3 + 1.5 \times \text{IQR}$$

Topic 13 & 14: Data Similarity & Dissimilarity Measures

1. Cosine Similarity (For Sparse Document/Text Vectors $\mathbf{x}, \mathbf{y}$):

$$\text{sim}(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\|_2 \|\mathbf{y}\|_2} = \frac{\sum_{i=1}^{n} x_i y_i}{\sqrt{\sum_{i=1}^{n} x_i^2}\sqrt{\sum_{i=1}^{n} y_i^2}}$$

2. Minkowski Distance Metric (L_p Norm):

$$d(\mathbf{x}, \mathbf{y}) = \left(\sum_{i=1}^{n} |x_i - y_i|^p \right)^{\frac{1}{p}}$$

- For $p = 1$: **Manhattan (L_1) Distance:** $d(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^{n} |x_i - y_i|$ - For $p = 2$: **Euclidean (L_2) Distance:** $d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$

🧠 M1 Active Recall & Exam Questions [UPLOADED PYQ]

Q1. [UPLOADED PYQ — IT426 Mid] Differentiate between Data Characterization and Data Discrimination with concrete examples. (8 Marks)

- **Data Characterization:** A summarization of the general characteristics or features of a target class of data. The output can be represented as pie charts, bar charts, multidimensional data cubes, or generalized relations.
Example: Characterizing customers who purchased items worth over ₹1,00,000 at an electronics store to find their average age, income, and city.
- **Data Discrimination:** A comparative analysis of the general features of the target class objects against one or more contrasting comparative classes.
Example: Comparing software engineers who remained at a company for over 5 years against those who left within 1 year to identify key turnover drivers.